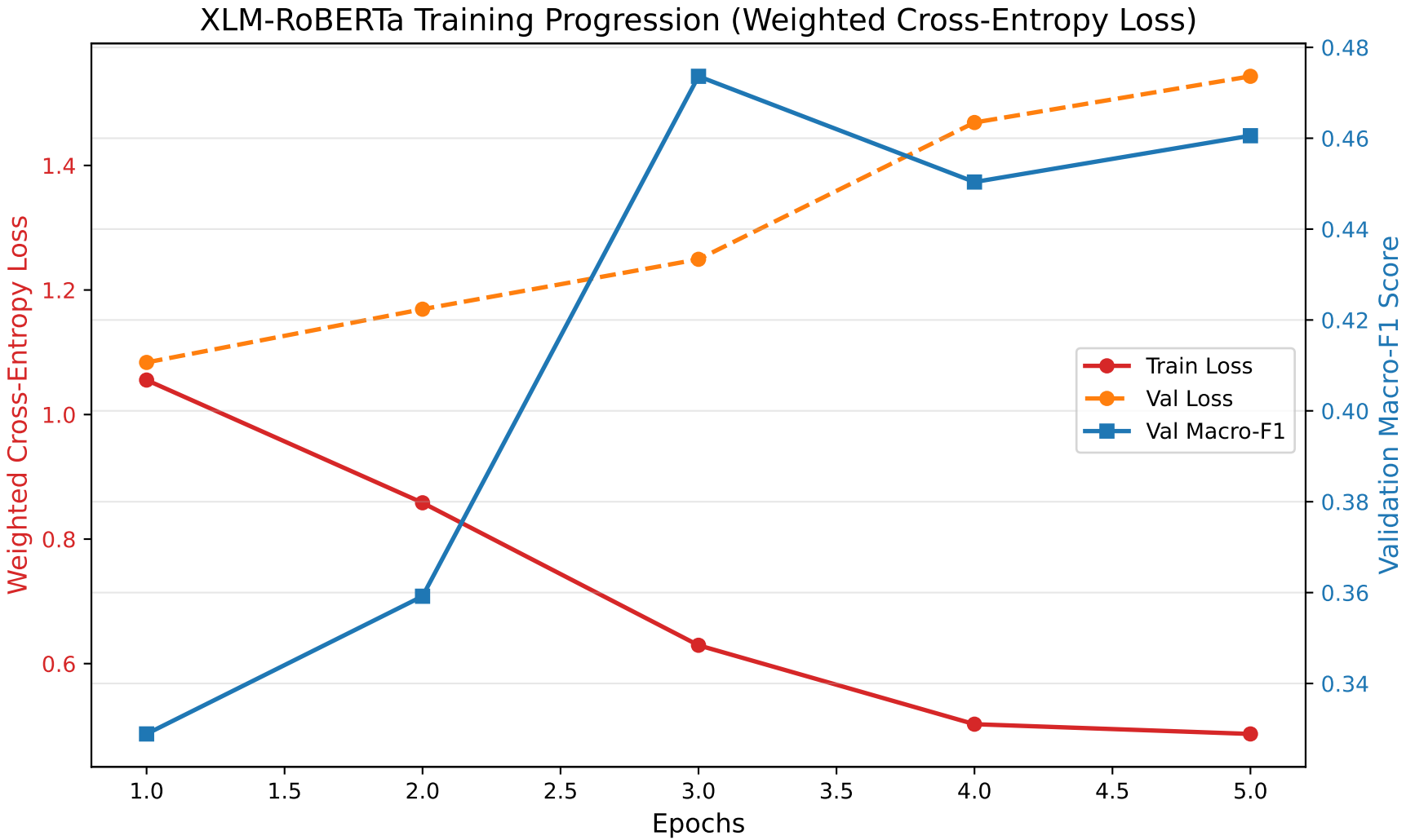
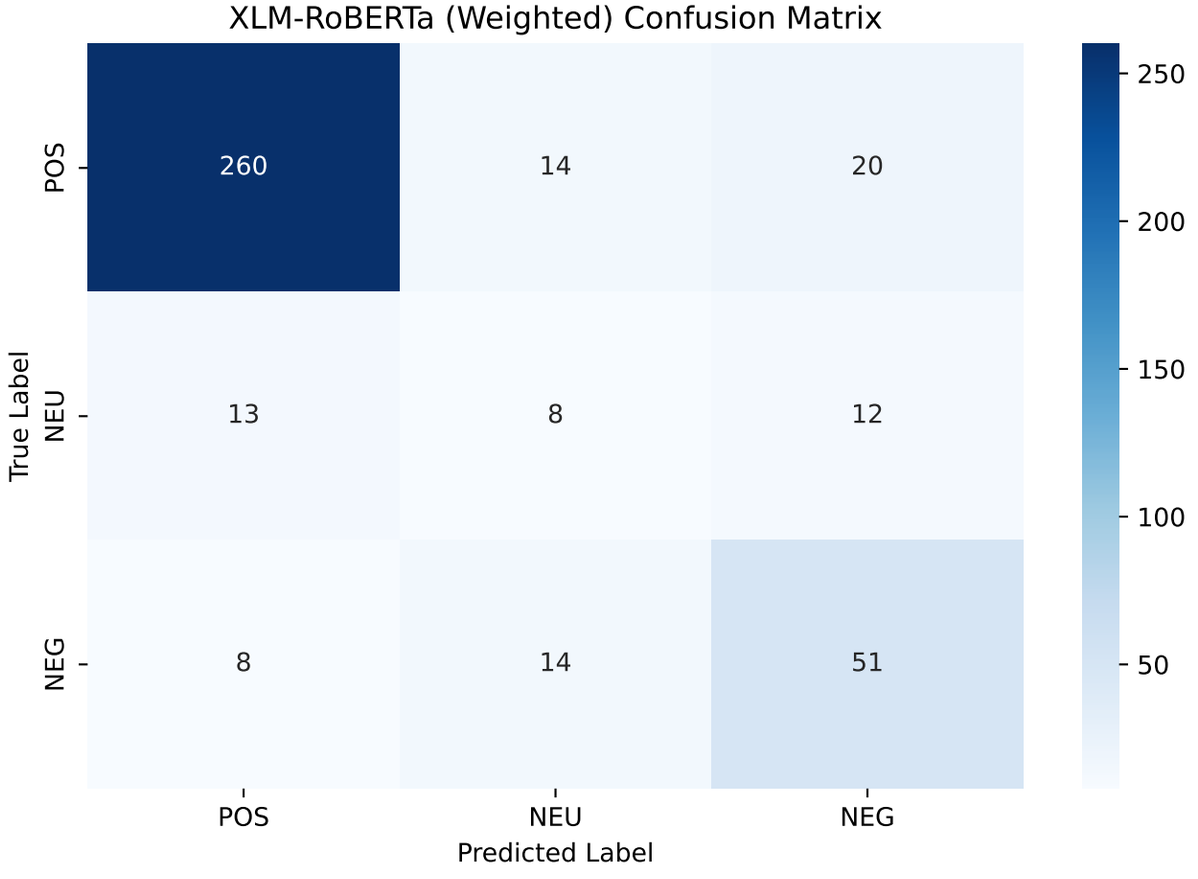
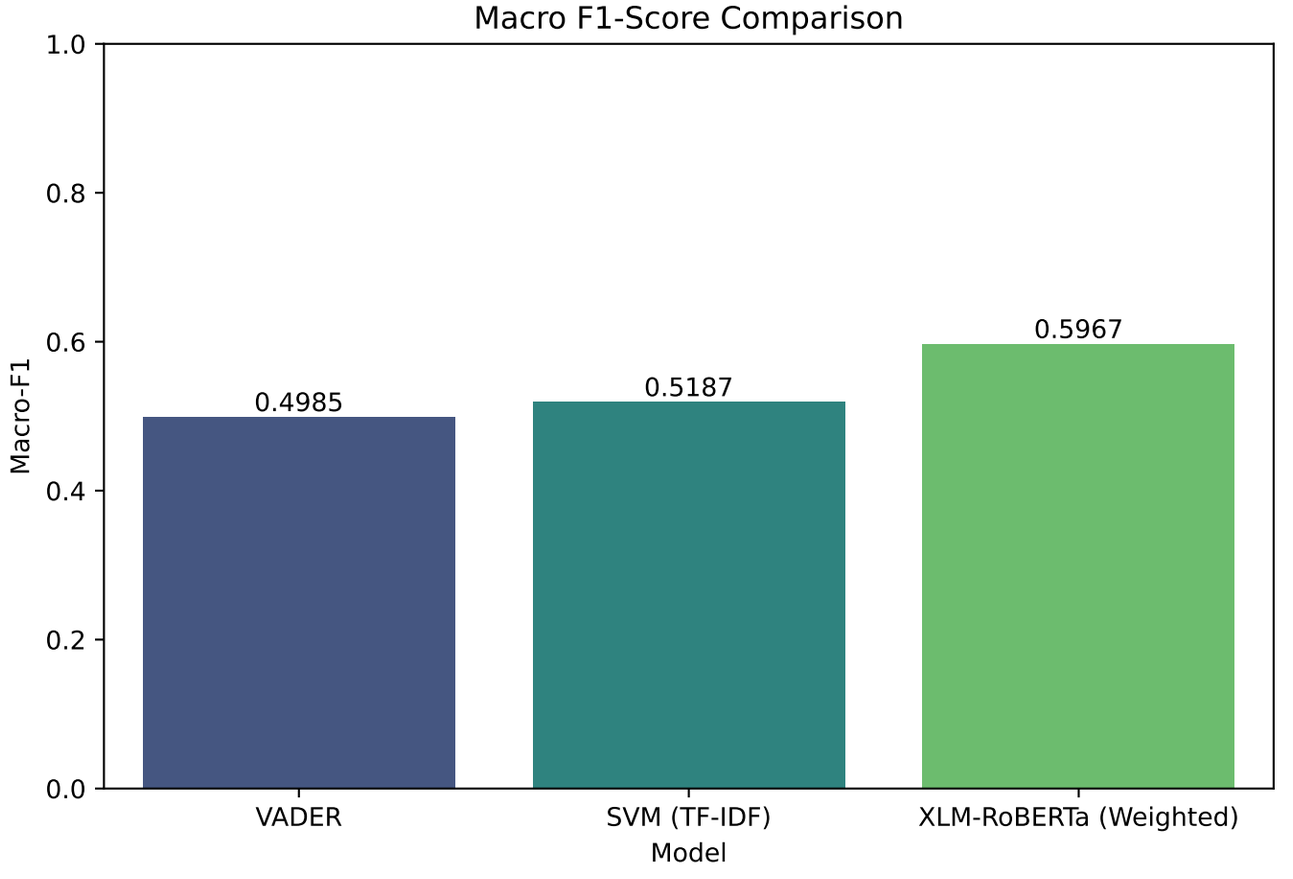

XLM-RoBERTa Training Progression (Weighted Cross-Entropy Loss)
Weighted Cross-Entropy Loss
Validation Macro-F1 Score
Epochs
Train Loss
Val Loss
Val Macro-F1
1.4
1.2
1.0
0.8
0.6
0.48
0.46
0.44
0.42
0.40
0.38
0.36
0.34
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


XLM-RoBERTa (Weighted) Confusion Matrix
True Label
POS
NEU
NEG
260
14
20
13
8
12
8
14
51
250
200
150
100
50
POS
NEU
NEG
Predicted Label


Macro F1-Score Comparison
Macro-F1
1.0
0.8
0.6
0.4
0.2
0.0
0.4985
0.5187
0.5967
VADER
SVM (TF-IDF)
XLM-RoBERTa (Weighted)
Model

PHASE 7/8: WEIGHTED LOSS FINE-TUNING & OUT-OF-SAMPLE EVALUATION REPORT

1. HYPERPARAMETER GRID SEARCH SUMMARY

Batch Size	Learning Rate	Epochs	Peak Val Macro-F1
16	0.00002	5	0.473630
32	0.00003	5	0.455092
16	0.00005	5	0.454746
16	0.00003	5	0.453645
32	0.00005	5	0.420121
32	0.00002	5	0.403029

2. FINAL BENCHMARK METRICS

Model	Accuracy	Precision (Macro)	Recall (Macro)	Macro-F1	F1 (POS)	F1 (NEU)	F1 (NEG)
VADER	0.6875	0.5685	0.5052	0.4985	0.8293	0.2034	0.4630
SVM (TF-IDF)	0.7300	0.5123	0.5290	0.5187	0.8502	0.1875	0.5185
XLM-RoBERTa (Weighted)	0.7975	0.5873	0.6085	0.5967	0.9043	0.2319	0.6538

3. MCNEMAR'S STATISTICAL TEST (alpha=0.05)

Baseline	XLM-R Correct (b)	Base Correct (c)	Chi-Square	p-value
VADER Lexicon	76	32	17.9259	2.2967e-05
SVM (TF-IDF)	48	21	10.5652	1.1524e-03